

Project Proposal

Bicycle Traffic Monitoring and Forecasting

Group 16

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Project Idea

In this project, we want to build an interactive dashboard for monitoring bicycle traffic at counting stations in Flanders and for producing short-term forecasts for selected stations. The data comes from the AWV bicycle counting system and includes monthly count files together with station metadata and direction metadata.

We deliberately define the project at the station level. The available data is collected at fixed counting points, so it is suitable for analysing traffic patterns at those locations, but not for describing the full road network in real time. For that reason, we see the main users of the dashboard as transport planners or infrastructure managers rather than individual cyclists.

Our aim is to answer two practical questions. First, how does bicycle traffic change across stations and over time? Second, can we make useful short-term predictions for a selected station based on its historical pattern?

Data

We will use the files in the AWV dataset:

- `data-YYYY-MM.csv`: monthly count data
- `sites.csv`: metadata about the stations, including coordinates and municipality
- `richtingen.csv`: metadata about the counting directions

From a first inspection of the data, the dataset contains around 151 stations and is mostly recorded at 15-minute intervals. The monthly files are fairly large, so an important first step will be to preprocess and aggregate the data before building the dashboard.

For the count files, we will use the variables `site ID`, `richting`, `type`, `van`, `tot`, and `aantal`. Since the dataset can include several object types, we will focus on FIETSERS only. We will then merge the count data with `sites.csv` and `richtingen.csv`.

Planned Method

Data cleaning and aggregation

We will first build a preprocessing pipeline in Python. The main steps will be:

- adding column names to the raw CSV files
- parsing the time variables
- filtering the data to bicycle counts only
- merging the monthly files into one dataset
- joining the count data with the station and direction metadata
- aggregating the counts from 15-minute level to hourly level

Because many stations have separate IN and OUT records, we will also create a station-level total count by combining both directions where appropriate. This will make the monitoring view and the forecasting task easier to interpret.

Exploratory analysis

Before building the forecasting model, we want to understand the main patterns in the data. In particular, we plan to study:

- differences in traffic intensity across stations
- daily and weekly traffic patterns
- weekday versus weekend differences
- seasonal variation across the year
- potential missing values or inactive periods

This step is important because it will show which stations are stable enough for forecasting and which temporal patterns are strongest.

Short-term forecasting

Our forecasting goal is modest: for a selected station, predict bicycle traffic for the next 24 hours or next 7 days at hourly level. We do not aim to make long-term forecasts or network-wide predictions.

We plan to compare:

- a simple seasonal naive baseline
- a Prophet model for short-term station-level forecasting

The baseline is important because it gives us a realistic reference point. If time permits, we may test one additional model such as SARIMAX, but our main comparison will be between a simple baseline and Prophet. Model performance will be evaluated with a time-based train-test split and metrics such as MAE, RMSE, and WAPE.

We may also add simple calendar features such as weekday/weekend and holidays. If time allows and suitable data is available, we may explore weather as an extension, but this is not essential for the core project.

Dashboard

The final product will be a Python Shiny dashboard. We currently imagine three main views:

- a map view showing the counting stations
- a station detail view with historical trends and summary plots
- a forecasting view showing future hourly counts for a selected station

In the map view, the user will be able to compare stations by traffic level. In the station detail view, the user will be able to inspect patterns such as peak hours or seasonal changes. In the forecasting view, the user will see the short-term prediction together with recent historical data.

To keep the app responsive, we will not load all raw CSV files directly in the dashboard. Instead, we will preprocess the data beforehand and store cleaned and aggregated tables in a more efficient format.

Tools

We plan to use:

- Python
- pandas or polars for preprocessing
- Plotly for interactive visualisation
- Python Shiny for the dashboard
- Prophet and statsmodels for forecasting
- Parquet or DuckDB for storing processed data

Expected Result

We expect to deliver a working dashboard, a cleaned station-level dataset, and a comparison of short-term forecasting approaches. We hope the project will show which stations have strong and predictable traffic patterns, and where short-term forecasting works reasonably well.

At the same time, we are aware of that the dataset is station-based, not full-network data, and it is updated monthly rather than in real time. Because of this, our project should be seen as a monitoring and planning tool, not as a live routing tool for cyclists.